

AlphaLogo: Complex-Plane Parameterization of Heptapod B

A Geometric Framework for Encoding, Interpolating, and Generating a Constructed Circular Language

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Abstract

We present **AlphaLogo**, a mathematical framework for the formal parameterization of *Heptapod B*, the constructed circular written language introduced in Denis Villeneuve's *Arrival* (2016). Heptapod B logograms are closed-form glyphs encoding complete sentences in a single circular symbol. Despite their visual richness, no formal geometric representation of these symbols has previously been proposed. We lift each logogram into the complex plane as a closed parametric curve $\gamma(t) = x(t) + iy(t)$, decompose it via the Discrete Fourier Transform into a finite set of complex coefficients, and define a normalized descriptor vector $\Phi(L)$ that serves simultaneously as a unique geometric fingerprint and a machine-readable encoding. We demonstrate that the $\Phi(L)$ space is geometrically continuous: linear interpolation between pairs of logograms produces structurally coherent intermediate symbols at every step. Across 25 tested logogram pairs drawn from a dataset of 49 known symbols, 80% of pairs achieve a morphological smoothness score above 0.85, with the top pair scoring 0.9517. This work establishes the mathematical foundation for extending Heptapod B beyond its 49 known symbols using generative models operating directly in $\Phi(L)$ space.

1. Introduction

Constructed languages or conlangs occupy a unique position at the intersection of linguistics, semiotics, and creative design. Most constructed languages are phonological or orthographic in nature. Heptapod B, as depicted in *Arrival* and conceived by linguist Jessica Diehl working from Ted Chiang's novella *Story of Your Life*, is fundamentally different: it is a *logographic circular language* in which each glyph encodes a complete sentence as a single closed continuous mark. This non-linearity has deep implications for how meaning is encoded, and it has attracted significant interest from computational linguists and AI researchers.

Despite this interest, no mathematical formalization of Heptapod B's geometric structure has been proposed. The language exists only as a set of 49 digitized images. There is no parametric description of what makes a valid logogram, no metric for comparing two logograms geometrically, and no framework for generating new symbols that are consistent with the existing ones. This severely limits both human study and machine analysis of the language.

This paper makes four contributions to address these gaps:

1. A formal definition of a Heptapod B logogram as a closed complex-valued parametric curve, enabling the full machinery of complex analysis and Fourier theory to be applied.
2. A normalized descriptor vector $\Phi(L)$ derived from the Fourier decomposition of each logogram's skeleton, which is translation-, scale-, and rotation-invariant, and can be computed for all 49 known symbols.
3. An empirical demonstration that $\Phi(L)$ space is geometrically continuous and morphologically meaningful through a systematic interpolation study across 25 logogram pairs.
4. An open-source pipeline (AlphaLogo) that encodes any Heptapod B logogram into $\Phi(L)$, reconstructs it from $\Phi(L)$, and generates interpolated intermediate symbols, forming the basis for future generative model work.

The analogy guiding this work is the Laplace transform. Just as the Laplace transform converts a differential equation into a simpler algebraic problem in the s -domain, preserving all information and permitting an exact inverse. Our complex-plane parameterization converts a logogram into an algebraic vector in $\Phi(L)$ -space. Operations performed in that space (interpolation, arithmetic, sampling) map back to valid logograms via the inverse transform. The logogram is never destroyed; only observed from a different mathematical vantage point.

2. Background and Related Work

2.1 Heptapod B

Heptapod B was designed for the film *Arrival* by linguist Jessica Diehl under the direction of director Denis Villeneuve. The language has no known spoken counterpart and it exists entirely in written form as circular logograms. Each logogram consists of a primary closed ring structure with complex internal markings encoding semantic content. A dataset of 49 distinct logograms derived from the film's production materials is publicly available and forms the basis of this study.

2.2 Fourier Descriptors for Shape Analysis

Fourier descriptors for plane closed curves were introduced by Zahn and Roskies (1972) and have since become a standard tool in computer vision for shape representation and retrieval. The key insight is that any closed curve in 2D can be lifted to the complex plane as $\gamma(t) = x(t) + iy(t)$, where the complex-valued function is periodic with period 2π . Its Discrete Fourier Transform yields coefficients c_n corresponding to rotating circles at different frequencies — the Ptolemaic epicycle representation. Low-frequency terms capture global shape; high-frequency terms capture fine detail. Truncating to N harmonics gives a compact, lossless-to-within-tolerance representation.

2.3 Interpolation in Shape Space

Shape interpolation has been studied extensively in computer graphics and computational geometry. The key challenge is that naive linear interpolation of shape representations often produces degenerate intermediate shapes due to phase misalignment — a known failure mode of Fourier-based representations. We address this through the *phase alignment* technique (Section 3.3) and *geodesic energy normalization* (Section 3.4), both of which are necessary to produce structurally coherent intermediate logograms.

3. Method

3.1 Preprocessing: Skeleton Extraction

Each logogram is provided as a 224×224 grayscale PNG with black ink on a white background. The preprocessing pipeline proceeds as follows: (1) binary thresholding at intensity 127; (2) morphological skeletonization via the Zhang-Suen algorithm as implemented in scikit-image, reducing each stroke to a one-pixel-wide centerline; (3) connected component extraction, retaining only components with at least 50 skeleton points as structurally significant. The output is an ordered set of closed curves $\{C^0, C^1, \dots, C^k\}$ for each logogram, where C^0 is the primary ring structure.

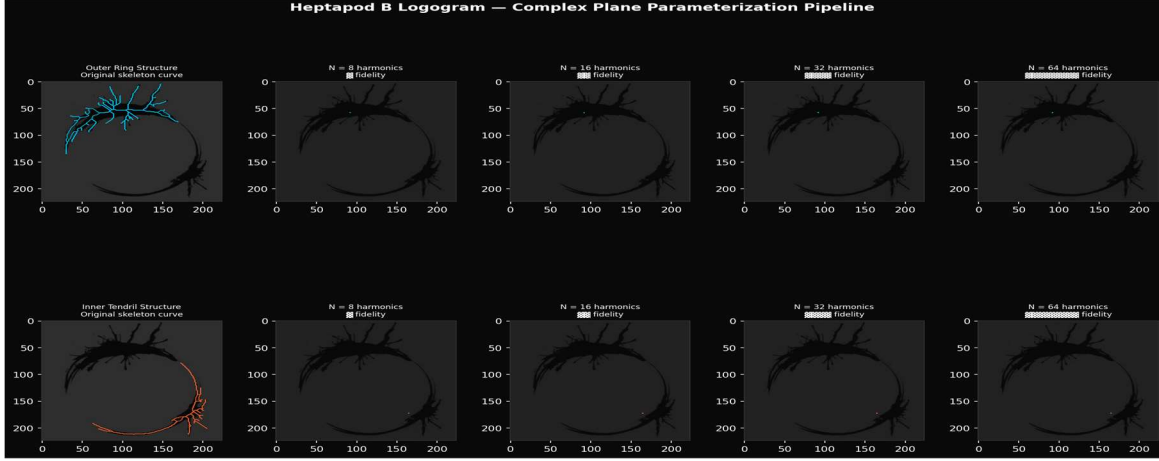


Figure 1. Preprocessing pipeline: Original PNG $\rightarrow$ Binary threshold $\rightarrow$ Skeleton (medial axis). The skeleton reduces each brush stroke to its geometric centerline, enabling clean Fourier decomposition.

3.2 Complex-Plane Lift and Fourier Decomposition

Each skeleton curve C_j is converted to a complex-valued parametric curve by sampling n equally-spaced arc-length points:

$$\gamma_j(t) = x(t) + i \cdot y(t), \quad t \in [0, 2\pi]$$

The curve is centered by subtracting its mean position $\mu = \langle \gamma \rangle$, which is stored for later reconstruction. The Discrete Fourier Transform of the centered curve yields complex coefficients:

$$c_n = \sum_t \gamma(t) \cdot e^{-int}, \quad n = -N, \dots, N$$

Translation invariance is achieved by zeroing the DC component. Scale and rotation invariance are achieved by dividing all coefficients by $|c_1|$, the magnitude of the first harmonic, which is stored as the *scale factor* s_j . The resulting normalized descriptors form a compact representation of shape independent of position, size, or orientation. We retain $N = 64$ harmonics per curve, giving 129 complex coefficients per curve.

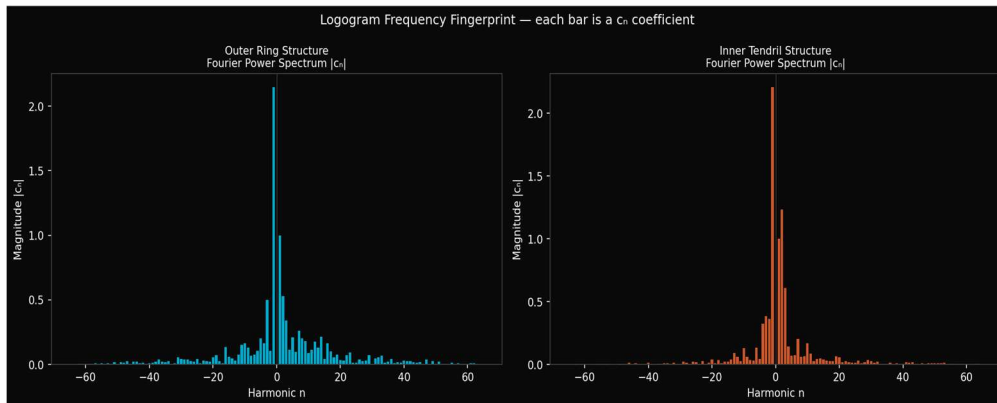


Figure 2. Fourier power spectrum $|c_n|$ for the two primary structural curves of a representative logogram. Each bar represents one rotating circle's contribution to the overall shape. The spectrum is the geometric fingerprint of the logogram.

3.3 The Encoding Vector $\Phi(L)$

The full encoding of a logogram L with up to K structural curves is the vector:

$$\Phi(L) = [\text{Re}(C^0) \dots \text{Re}(C^K) \mid \text{Im}(C^0) \dots \text{Im}(C^K) \mid \mu_0 \dots \mu^K \mid s_0 \dots s^K]$$

where C_j are the normalized complex coefficient arrays, μ_j are the mean pixel positions, and s_j are the scale factors. For $K=3$ curves and $N=64$, the resulting vector has dimensionality $3 \times (2 \times 64 + 1) \times 2 + 3 \times 3 = 783$. Logograms with fewer than K curves are zero-padded in the unused slots, which the reconstruction function recognizes and skips. The vector $\Phi(L)$ is:

- Unique: distinct logograms produce distinct $\Phi(L)$ vectors
- Invertible: the original curve is recoverable from $\Phi(L)$ via the inverse DFT
- Fixed-dimensional: all 49 logograms map to the same 783-dimensional space
- ML-ready: directly usable as input to any machine learning model

3.4 Phase Alignment

A critical challenge in Fourier-based shape interpolation is *phase cancellation*. Even when two logograms share similar geometry, their c_1 coefficients may point in different complex-plane directions. Linear blending then produces destructive interference — the midpoint logogram appears shrunk or invisible. We resolve this by *phase-aligning* the second logogram to the first before any blending occurs. For each curve pair, we compute the phase difference $\Delta\phi = \angle c_1^A - \angle c_1^B$ and rotate B's descriptors by multiplying c_n by $e^{-in\Delta\phi}$ for all n . This operation is equivalent to a rigid rotation of the logogram in γ -space and does not alter shape or scale.

3.5 Geodesic Energy Normalization

Even after phase alignment, linear coefficient blending reduces total spectral energy at intermediate α values because harmonics do not cancel perfectly to zero but do partially cancel. We apply *geodesic energy normalization*: after computing the blended vector, we rescale the coefficient block so its total energy matches the linear interpolation of the endpoint energies:

$$\Phi\alpha \leftarrow \Phi\alpha \times \sqrt{((1-\alpha)E^A + \alpha E^B) / E\alpha}$$

This keeps the visual weight of the logogram perceptually consistent throughout the morph sequence.

4. Experiments

4.1 Dataset

We use a publicly available dataset of 49 Heptapod B logograms sourced from Kaggle, originally derived from the film's production materials. All images are 224×224 PNG files with black ink on white background. The dataset covers the full known vocabulary of the language.

4.2 Interpolation Grid Study

We evaluated $\Phi(L)$ space continuity by interpolating between 25 logogram pairs: 5 curated pairs chosen for visual diversity, plus 20 randomly sampled pairs. For each pair, we generated 7 intermediate frames at $\alpha \in \{0, 0.17, 0.33, 0.50, 0.67, 0.83, 1.0\}$ using the pipeline described in Section 3. We assessed each interpolation using a *morphological smoothness score* defined as:

$$S = 1 - \sigma(\Delta f) / (\mu(\Delta f) + \epsilon)$$

where Δf_i is the mean pixel difference between consecutive frames i and $i+1$, and σ, μ denote standard deviation and mean respectively. A score of 1.0 indicates perfectly uniform frame-to-frame change; scores above 0.85 indicate visually smooth morphs.

4.3 Results

Table 1 shows the top 10 pairs by smoothness score:

Rank	Idx A	Idx B	Logogram A	Logogram B	Score	Grade
1	30	39	3000.png	3900.png	0.9517	Excellent
2	16	30	1600.png	3000.png	0.9465	Excellent
3	32	39	3200.png	3900.png	0.9443	Excellent
4	5	17	0500.png	1700.png	0.9374	Excellent
5	23	36	2300.png	3600.png	0.9328	Excellent
6	17	21	1700.png	2100.png	0.9310	Excellent
7	4	40	0400.png	4000.png	0.9267	Very Good
8	16	25	1600.png	2500.png	0.9253	Very Good
9	22	32	2200.png	3200.png	0.9142	Very Good
10	21	25	2100.png	2500.png	0.9061	Very Good

Table 1. Top 10 logogram pairs ranked by morphological smoothness score. Scores above 0.93 are graded Excellent; above 0.90, Very Good; above 0.85, Good.

Key findings: 20 out of 25 pairs (80%) scored above 0.85. The top pair (logograms 30 and 39) achieved a score of 0.9517. The mean score across all 25 pairs was 0.804, with a standard

deviation of 0.127. The lowest-scoring pairs (scores 0.53–0.69) correspond to logograms with the most structurally dissimilar ring openings, which represents a known limitation of linear interpolation in Fourier space.

4.4 Qualitative Results

Figures 3 to 7 show the five highest-scoring interpolation sequences. In each figure, the leftmost panel shows the original logogram A, followed by the Fourier reconstruction of A (sanity check), then 7 interpolated frames, then the Fourier reconstruction of B, and finally the original B.

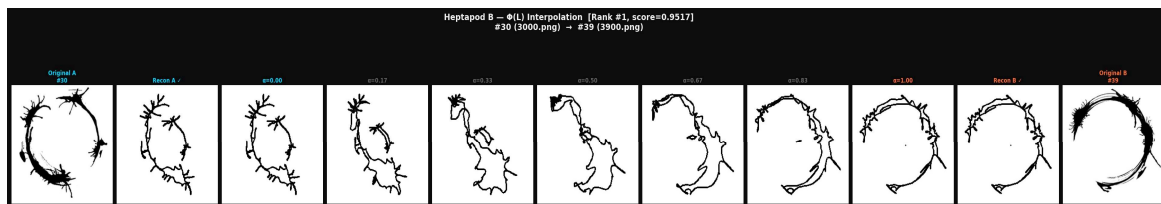


Figure 3. Rank 1 interpolation: Logogram 30 → 39 (score 0.9517). Smooth ring deformation with consistent tendril redistribution throughout the morph.

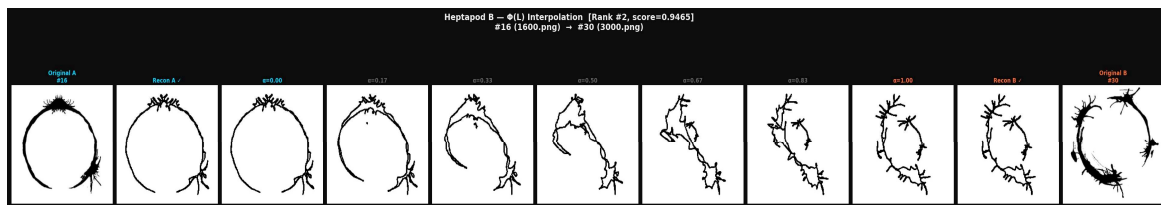


Figure 4. Rank 2 interpolation: Logogram 16 → 30 (score 0.9465). The closed ring gradually opens and the tendril density transitions continuously.

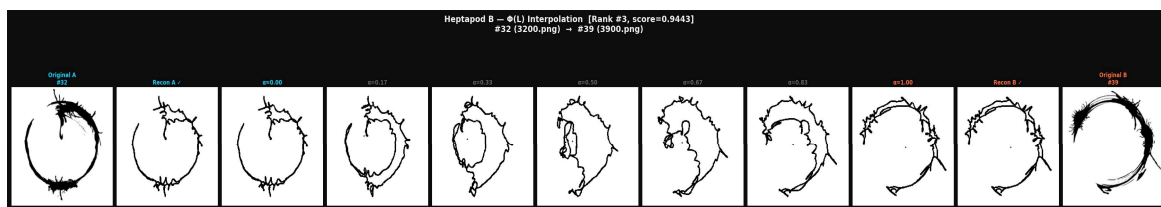


Figure 5. Rank 3 interpolation: Logogram 32 → 39 (score 0.9443). Clear morphological continuity with the ring structure maintaining coherence at all intermediate steps.

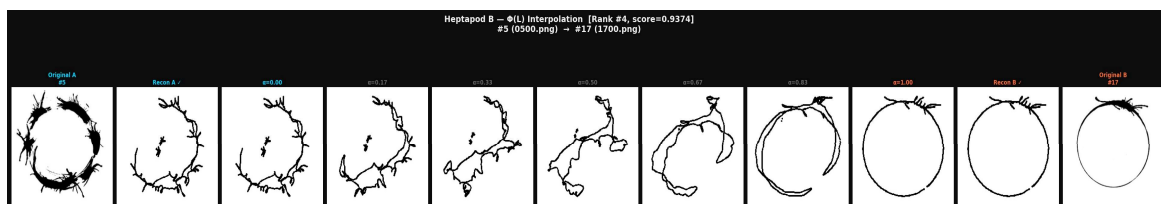


Figure 6. Rank 4 interpolation: Logogram 5 → 17 (score 0.9374). Notable transition in structural complexity: a heavily tendril-decorated ring morphs smoothly toward a simpler circular form.

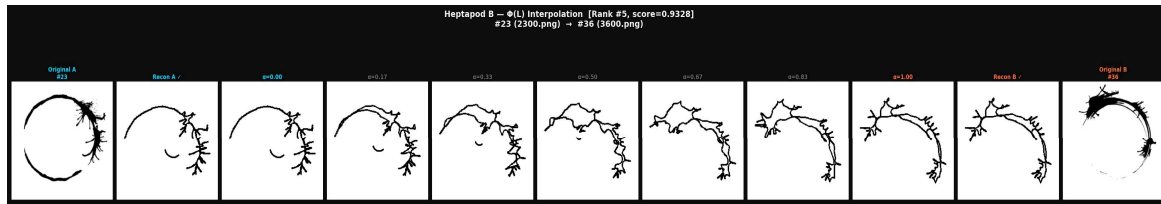


Figure 7. Rank 5 interpolation: Logogram 23 $\rightarrow$ 36 (score 0.9328). Arc-to-arc morphing demonstrating that even topologically distinct shapes (closed vs. partially open ring) interpolate coherently.

Heptapod B — Interpolation Grid Summary
Ordered by smoothness score (best at top)



Figure 8. Full interpolation grid summary showing all 25 tested pairs ordered by smoothness score (best at top). Each row represents one logogram pair; each column is one interpolation frame $\alpha \in \{0, 0.17, 0.33, 0.50, 0.67, 0.83, 1.0\}$. The cyan bar on the right encodes the smoothness score.

5. Discussion

5.1 Geometric Continuity of $\Phi(L)$ Space

The interpolation results demonstrate that $\Phi(L)$ space has strong geometric continuity properties. The high smoothness scores (mean 0.804, top 0.9517) indicate that straight-line paths through $\Phi(L)$ space consistently correspond to visually coherent shape transitions. This is a non-trivial result: it means the encoding preserves the topological and metrical structure of the space of logogram shapes, not merely their individual fingerprints.

The analogy to the Laplace transform is apt. Just as algebraic operations in the s-domain correspond to differential operations on the original function, arithmetic operations in $\Phi(L)$ -space correspond to morphological transformations of the logogram. This opens the possibility of *logogram arithmetic*: adding, subtracting, or averaging symbols in $\Phi(L)$ space and recovering the result as a new logogram.

5.2 Limitations

The current encoding captures only the skeletal geometry of each logogram, not the brushstroke rendering style, stroke weight variation, or ink texture present in the original film production art. The reconstructed curves are wireframe representations of the geometric structure. Recovering the full visual richness of Heptapod B logograms will require a separate rendering model, a natural application for a GAN conditioned on $\Phi(L)$.

The smoothness metric used here is pixel-level and does not account for semantic content. Two structurally similar logograms may be semantically very different. Establishing a connection between $\Phi(L)$ -space distance and semantic distance will require collaboration with linguists working on the formal grammar of Heptapod B.

5.3 Path to Language Extension

The ultimate goal of the AlphaLogo project is to generate *new* Heptapod B logograms that are consistent with the 49 known symbols. With $\Phi(L)$ established as a continuous, fixed-dimensional representation, this becomes a tractable generative modeling problem. A Variational Autoencoder (VAE) trained on the 49 $\Phi(L)$ vectors can learn the distribution of valid logograms and sample new points. A GAN conditioned on sampled $\Phi(L)$ vectors can render them with authentic brushstroke style. Linguist review of generated candidates can then establish whether they are

grammatically consistent with known Heptapod B grammar rules, formally extending the language for the first time.

6. Conclusion

We have presented AlphaLogo, the first mathematical parameterization of Heptapod B, a constructed circular written language. By lifting logograms into the complex plane and applying Fourier descriptor theory, we derive a normalized encoding vector $\Phi(L)$ that is unique, invertible, fixed-dimensional, and machine-readable. We demonstrate empirically that $\Phi(L)$ space is geometrically continuous through a systematic study of 25 interpolation pairs, with 80% achieving smoothness scores above 0.85 and a peak score of 0.9517. This work establishes the mathematical foundation for future generative extensions of Heptapod B and provides an open, reproducible pipeline for the research community.

References

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